Report - FINAL PROJECT

STAT 480 Big Data Analytics

University of Illinois Urbana-Champaign

Airbnb Market Analysis Across Major U.S. Cities

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INTRODUCTION

This project aims to analyze and model Airbnb performance across four major U.S. markets: New York City, Los Angeles, Dallas, and Chicago. Using publicly available Inside Airbnb datasets (listings, calendar, reviews, and neighborhood data), the focus is on understanding determinants of pricing and occupancy as well as identifying meaningful market patterns.
The analysis seeks to answer four key research questions:

1. Are there performance-based segments of listings that share similar pricing levels, quality signals, and booking patterns? ⇒ SEGMENTATION.
2. What property, host, and neighborhood characteristics most strongly influence nightly prices? ⇒ PRICE MODEL.
3. What factors predict whether a listing achieves high occupancy rates? ⇒ BOOKING MODEL.
4. Does Superhost status have a causal effect on occupancy, controlling for other listing characteristics? ⇒ TREATMENT EFFECT.

These questions combine predictive and causal inference approaches using methods such as K-means clustering, LASSO regression, logistic models, and treatment effect estimation (Double LASSO).

DATA PREPROCESSING - Diego Luis de Mazarredo

We work with Inside Airbnb data for four cities (New York, Los Angeles, Chicago, Dallas). Each city provides two large CSV files: a *listings* file with host, property, and review attributes, and a *calendar* file with day-level prices and availability. Our preprocessing pipeline transforms these raw files into two modeling-ready objects: a unified listing table (**all_listings**) and a high-dimensional, sparse amenities matrix (**amenities_sparse**)

Cleaning calendar data:

For each city, the calendar dataset is standardized and filtered to retain only useful features. We convert *listing_id* to character for consistent merging, transform dates into proper Date objects, and derive time-based variables such as month, weekday, and weekend indicators. Price fields are cleaned by removing symbols (for instance,. $120.00) and converting them to numeric. Redundant metadata (e.g., scraping timestamps, license fields) is removed to reduce memory usage and noise. The result is a lightweight per-city calendar table suitable for aggregation.

Cleaning listings data:

Listing-level datasets undergo type normalization and variable selection. IDs (*id*, *host_id*) are converted to character, and key numeric fields (*price*, *minimum/maximum nights*) are sanitized to numeric. Categorical variables such as *room_type* are cast as factors. A simple amenity summary variable (*amenity_count*) is constructed by counting elements in the raw amenity string. Long text fields, URLs, and cosmetic attributes are removed to keep only relevant structural, host, and review features.

Aggregating calendar information into listing-level features:

To capture booking dynamics, we compute listing-level summaries from the cleaned calendar data:

- **Occupancy rate:** proportion of days marked unavailable (“f”), interpreted as booked.
- **Average available price:** mean daily price on days marked available (“t”).

These aggregates are merged back into the listings tables, creating a richer feature set that connects availability patterns with listing characteristics.

Sparse amenities matrix:

Amenities are high-dimensional and sparse: each listing has only a small subset of many possible amenities. To represent this efficiently we create a sparse dummy matrix for each city using sparse.model.matrix:

- one binary column per amenity (e.g. amenityWifi, amenityPool),
- 1 if the listing has that amenity, 0 otherwise, stored in a sparse representation.

These per-city matrices are later stacked into a single **amenities_sparse** matrix aligned with *all_listings*. This structure is designed for use in high-dimensional models such as LASSO, where storing a dense dummy matrix would be too memory-intensive.

Combining cities & creating modeling variables:

All cleaned listing tables are merged and tagged with a city indicator. Price is log-transformed (*log_price = log(price)*) to reduce skewness and stabilize variance. Boolean Airbnb flags stored as "t/f" strings are converted to 0/1 indicators. For occupancy modeling, we define *high_occupancy = 1* when occupancy rate exceeds 0.8.

Handling missing values:

We implement a targeted missing-data strategy. Listings with missing price are dropped; review-related fields are set to zero for listings without reviews and median-imputed when partial information is missing. Small-NA numeric variables (e.g., host listing counts) are median-imputed. Beds, bedrooms, and bathrooms are imputed within city- and room-type groups. Missing average available prices are filled using either the nightly price or the global median. Residual categorical missingness is assigned to an "Unknown" category when appropriate.

At the end of preprocessing we save two key objects: **'all_listings.rds'**, the cleaned, combined listing-level dataset with and **'amenities_sparse.rds'**, a large sparse matrix of amenity dummies aligned with all_listings. These objects allow us to start the modeling stage directly.

ANALYSIS OF KEY VARIABLES

This section presents the statistical analysis of the four key variables - price, occupancy, segmentation, and treatment effect - using predictive, unsupervised, and causal inference methods.

1. SEGMENTATION - Felipe Grima

This section uses unsupervised learning to discover patterns in Airbnb listings and determine whether naturally occurring market segments exist. We combine PCA for dimension reduction with K-means to produce interpretable groups. We perform segmentation in two stages: **amenity-based segmentation** (high-dimensional sparse amenities) and **segmentation based on price–quality–occupancy features.** These two steps together allow us to understand both what listings offer and how they perform.

CAN WE INFER MEANINGFUL SIMILARITIES BETWEEN LISTINGS BASED SOLELY ON THEIR AMENITY PROFILES?

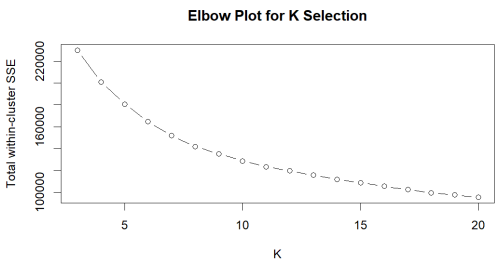
We apply PCA and clustering to the high-dimensional amenity space in order to uncover systematic patterns in how listings are equipped.

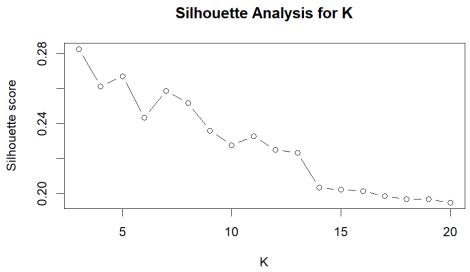
PCA

Given that the amenity matrix is high-dimensional and sparse, we compute PCA using the "irlba" algorithm, which efficiently approximates the leading principal components without forming the full covariance matrix.
Checking the variance explained by the first 20 principal components (PCs), we realize that the first 6 PCs explained ⅔ of total variance. Thus, seeking for interpretability, we chose that amount of PCs, meaning:

PC	Meaning	High value (positive loadings)	Low value (negative loadings)
PC1	Modern Comfort vs. Basic Interior Setup	Air conditioning, TV, washer, WiFi, lock on bedroom door	Toaster, dining table, wine glasses, cleaning products, freezer
PC2	Indoor Heating & Essentials vs. Outdoor/Shared Features	Heating, coffee maker, oven, stove, dryer	Coffee, cleaning products, outdoor dining area, central heating (shared)
PC3	Urban Self-Service Units vs. Home-Style Amenities	Essentials, hangers, street parking, self check-in, iron	Outdoor dining, AC, TV, garage/driveway parking, washer
PC4	Outdoor & Pet-Friendly vs. Indoor Personal-Care Focus	BBQ grill, outdoor dining, outdoor furniture, dishwasher, pets allowed	Cleaning products, clothing storage, shower gel, body soap, conditioner
PC5	Full Kitchen Functionality vs. Host Safety/Essentials	Dishwasher, oven, cooking basics, kitchen, dining table	Luggage dropoff, heating, shampoo, fire extinguisher, first aid kit
PC6	Practical/Security Amenities vs. Luxury Building Features	Free street parking, stove, dishes & silverware, fire extinguisher, security cameras	Conditioner, gym, TV, pets allowed, elevator

K-MEANS





Searching for optimal K for K-means, we choose K=8. Despite the absence of a sharp optimum from standard criteria, we select this number of clusters by balancing stability and interpretability.

With this, we can conclude the following clusters:

Cluster	PCA Profile	Segment Name	Description
1	Moderately modern, strong comfort & AC, average kitchen, positive PC6 (security)	**Urban Secure Studios**	Small units with decent comfort + strong safety features (cameras, extinguishers).
2	Mild indoor focus, light kitchen, fewer outdoor features	**Basic Indoor Units**	Minimal outdoor amenities, modest kitchen features, simple essentials.
3	Strong PC4 & PC3, outdoor & self-service heavy	**Outdoor Self-Service Homes**	Outdoor living amenities + self check-in + moderate comfort.
4	Mixed profile, balanced	**Balanced Amenities Homes**	Moderate in all PCs, no extreme amenity specialization.
5	High kitchen (PC5 neg), full cooking setups	**Full-Kitchen Apartments**	Strong cooking amenities, oven, dishwasher, kitchen-centric listings.

6	PC3/PC6 moderate, low outdoor	**Minimalist Practical Units**	Practical, self-service studios with few outdoor or luxury features.
7	High PC1 comfort, some outdoor	**Luxury Comfort Homes**	Very comfort-heavy units with strong indoor amenities.
8	Moderate comfort, slight outdoor penalty	**Indoor-Focused Micro-Homes**	Very compact units with limited outdoor features; some security and kitchen basics.

CAN WE IDENTIFY CLUSTERS OF LISTINGS WITH SIMILAR PRICE-QUALITY-OCCUPANCY PROFILES?

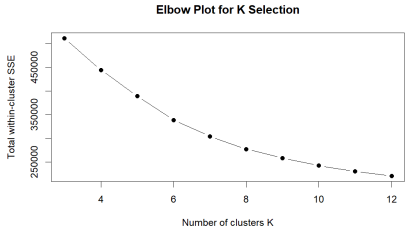
PCA

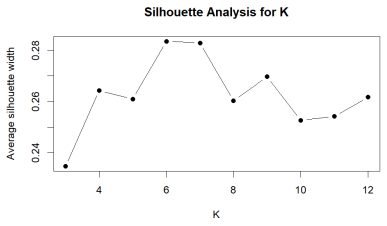
We select the following set of continuous and binary performance-related variables: log_price, occupancy_rate, number_of_reviews, reviews_per_month, review_scores_rating, amenity_count, accommodates, bedrooms, beds, bathrooms, host_is_superhost, host_listings_count, host_identity_verified, availability_365.

We perform PCA, using the standard "prcomp" implementation, in these listings dataset's variables. With this procedure, we obtain almost 80% of variance explained with the first 6 PCs:

PC	Meaning	High value means	Low value means
PC1	Price & size	High price, many beds, large listings	Budget/small listings
PC2	Reviews & amenities	High amenity_count, many reviews	Few reviews/amenities
PC3	Host professionalism	High host listings count, superhost	Casual hosts, small inventory
PC4	Occupancy intensity	High occupancy_rate	Low occupancy_rate
PC5	Rating & quality	High review scores	Lower review scores
PC6	Availability pattern	Low availability = more booked	High availability = less booked

K-MEANS





We choose K=6. Our Silhouette score clearly peaks at K = 6 and the Elbow plot stabilizes at K = 6. Thus, we obtain the following clusters (numerical summaries below are reported for descriptive purpose).

Cluster	Segment Name	Main Traits
1	**Urban Budget Micro-Studios**	Very cheap (avg price around $154), extremely low occupancy (0.11), small units (approx. 2.8 accommodates, 1.3 beds), few amenities, low reviews, mid-size hosts. Often low-demand city micro-units.

2	**Luxury Large-Group Homes**	Extremely expensive ($679), large homes (approx. 8.9 accommodates, 4.9 beds, 3.8 bedrooms), high amenities, moderately high reviews, higher occupancy (0.29). High-end villas / large family accommodations.
3	**Premium Apartments & Mid-Size Homes**	High price ($391), moderate occupancy (0.23), mid-size homes (around 1.5 bedrooms), strong amenities, average reviews. Typical modern full apartments or small houses.
4	**Hyper-Demand Budget Studios**	Low price ($163), very high occupancy (0.69), small units but very popular, mid-level amenities, good review scores, mid-size hosts. Most booked "cheap but good" listings.
5	**Superhost High-Quality Rentals**	Moderate price ($166), solid occupancy (0.37), strong amenity count (around 47), very high review volume (approx. 182), highest superhost rate (79%). High-quality, guest-favorite apartments.
6	**Corporate Mid-Range Units**	Mid-range price ($174), moderate-high occupancy (0.34), strong amenities, good reviews, very large host portfolios (around 21 listings). This means corporate/megahost-managed consistent units.

Across both amenity-based and performance-based analyses, the segmentation uncovers coherent business-relevant clusters:

- Amenity PCA reveals **systematic investment patterns** (comfort, outdoor living, cooking functionality, safety).
- Performance PCA captures **latent drivers of success**: size/price, ratings, host professionalism, occupancy intensity, and availability strategy.
- K-means clustering on these components identifies **distinct market segments** ranging from hyper-demand budget studios to luxury large-group homes and corporate-managed rentals.

This confirms that Airbnb listings **do indeed form interpretable market segments**, and each group displays a **different combination** of price level, occupancy behavior, host sophistication, and amenity richness. These segments can guide hosts in benchmarking their listing type and identifying improvement priorities.

2. PRICE MODEL - Adriana Zugasti

In this subsection we model the logarithm of the nightly price (log_price) for all Airbnb listings in our combined dataset (New York, Los Angeles, Dallas and Chicago). We use LASSO regression to handle the large number of correlated predictors and to obtain a sparse, interpretable model with good out-of-sample performance.

Data preparation and outcome:

We start from the cleaned all_listings table and keep only rows with a valid positive price and non-missing log_price. This removes observations with zero, negative or missing prices. To reduce the influence of extreme luxury listings, we cap the price distribution at the 99.5th percentile and drop the top 0.5% most expensive listings before fitting the model.

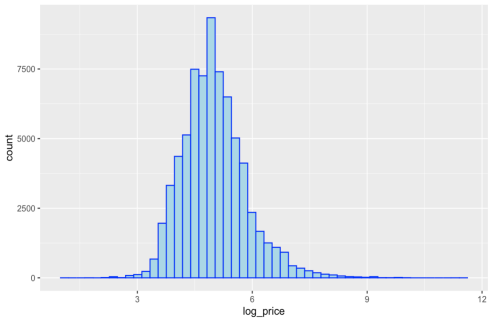
The response variable is

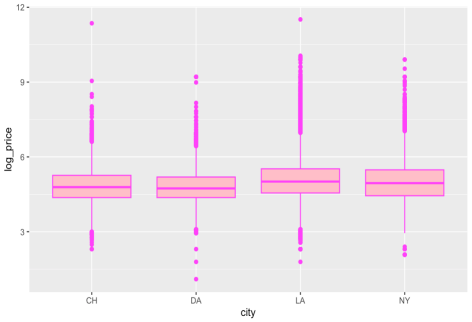
$$Y=log(price),$$

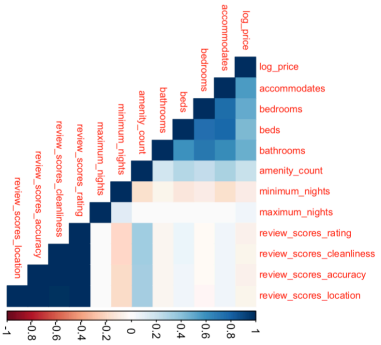
which makes the distribution closer to normal and allows us to interpret coefficients as approximate percentage changes in price.

Exploratory analysis:

This section intended to give us more information on the variables included to decide whether some were redundant or we could apply a cap to eliminate outliers, etc. To further refine our selection before inputting them into the model.







Histogram: The raw price is extremely right–skewed, while log_price is much more symmetric, so we model log_price as the response and trim the most extreme high prices to reduce the influence of outliers.

Box plots: We see clear level differences across cities (NY/LA more expensive than DA/CH), which motivates including city as a categorical predictor so these systematic market effects are explicitly controlled for.

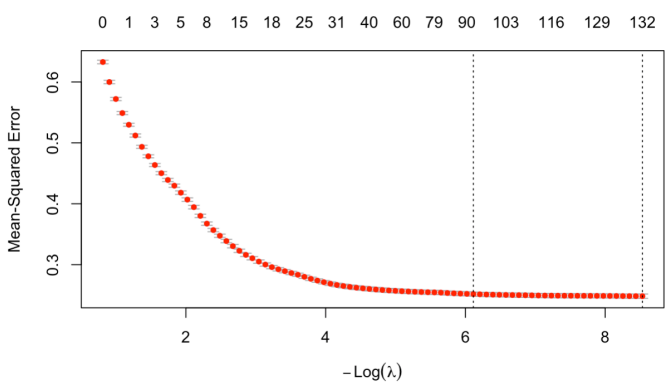
Correlation matrix: There is a strong positive relationship between size and log_price, so we include size variables (accommodates, bedrooms, bathrooms, beds) as key predictors and later consider simple non–linear terms to capture mild curvature. Size variables and review scores are highly correlated within their blocks, which justifies using LASSO to handle multicollinearity and let the penalty select only the most informative variables from each correlated group.

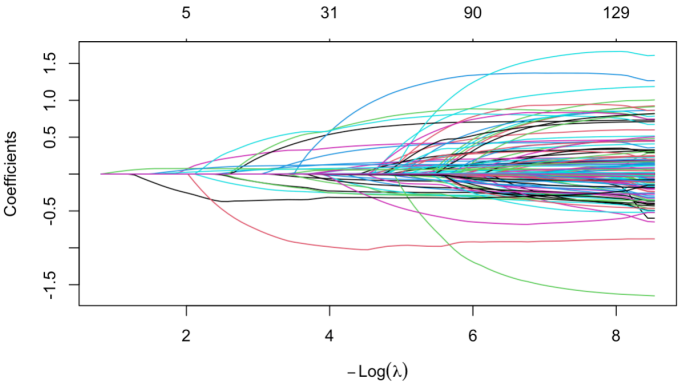
Predictor construction:

Because the exploratory plots showed strong effects of *city*, *location*, and *size*, we include *city*, *neighbourhood_group_cleansed, accommodates, bedrooms, beds, bathrooms* and *amenity_count* as core predictors. We also add key host and demand signals - *host_is_superhost, instant_bookable, number_of_reviews, reviews_per_month, and review_scores_rating* - since they plausibly affect perceived value. Given the high correlations inside the size and review blocks, we keep the set relatively compact and rely on the LASSO penalty to down–weight or drop redundant variables.

LASSO Price Model (version 1):

We first fit a standard LASSO regression of log_price on all selected predictors using cv.glmnet with 5-fold cross-validation and alpha = 1. We choose the penalty parameter via the 1-SE rule, using lambda.1se, which gives a model that is almost as accurate as the minimum-CV-error model but sparser and more stable.





On the test set, Model 1 attains MSE≈0.249, and an $R^2 \approx 0.62$, very close to the training R^2, indicating limited overfitting.

Top 20 features	Beta	% change
property_typePrivate room in resort	1.3376259	280.99876
property_typeShared room in villa	-1.3350136	-73.68454
property_typePrivate room in yurt	1.2912008	263.71514
property_typeEntire home/apt	0.9312087	153.75746
property_typeRoom in boutique hotel	0.8726306	139.31980
room_typeShared room	-0.8273286	-56.27843
property_typeRoom in hotel	0.8039661	123.43852
property_typeRoom in b&b	0.8004218	122.64798
property_typeRoom in serviced apartment	0.7512747	111.97002
property_typeEntire villa	0.7046654	102.31695
room_typeHotel room	-0.6475658	-47.66819
property_typeHouseboat	0.5965376	81.58208
property_typeEntire resort	0.5320289	70.23828
property_typeBoat	0.4881172	62.92457
property_typePrivate room in boat	0.4790174	61.44873
property_typeFloor	0.4551854	57.64657
property_typeShared room in b&b	0.4497857	56.79761
property_typeShared room in home	-0.4291762	-34.89548
property_typeCave	0.4248383	52.93432
neighbourhood_group_cleansedManhattan	0.4202367	52.23218

We also looked at the top 20 beta coefficients to determine which variables have the largest impact on predicted *log_price* relative to the baseline. A positive β / perc_change (e.g. "Private room in resort", +281%) means that type is estimated to be more expensive than the baseline, while a negative value (e.g. "Shared room in villa", −74%) indicates a cheaper listing, holding other variables fixed.

From these findings we reached 3 main conclusions which aligned with our exploratory analysis.

1. **Property type / room type** are the main drivers of price, with entire homes/apartments and hotel-like properties having large positive effects, and shared rooms receiving strong discounts relative to the baseline.
2. **Location effects** are important: for example, listings in Manhattan or other central areas are substantially more expensive than the reference neighbourhood group.
3. **Size and amenities** (accommodates, bathrooms, amenity_count) retain moderate positive coefficients, while many review-related and highly correlated variables are either heavily shrunk or dropped.

These results highlight that these variables are the dominant determinants of price, while many secondary variables (e.g., some review metrics or availability measures) are shrunk toward zero.

Extended LASSO Model (version 2):

We estimated a second LASSO model adding simple non-linear terms (e.g. accommodates², bathrooms²) and interactions between city and room_type to let room-type effects differ by market. Using the same CV procedure and 1-SE rule, the test **MSE improved to 0.240** and **R^2 to 0.63**. Thus, richer structure gives a small predictive gain but at the cost of a more complex and less interpretable model, so we mainly treat Model 2 as a robustness check.

Model evaluation and limitations:



Train and test performance are very similar and the predicted vs actual log_price plots cluster around the 45° line, suggesting limited overfitting, with larger residuals mainly for very high-priced listings. However, the models remain linear in the chosen features and cannot capture more complex patterns we did not code explicitly. In addition, important drivers such as detailed location, interior quality, or host behaviour are unobserved, and the cross-sectional nature of the data means we do not account for seasonality or dynamic pricing, which likely contributes to remaining error - especially in the luxury segment.

3. BOOKING MODEL - Diego Luis de Mazarredo

We're going to build a **booking model** that predicts whether an Airbnb listing reaches **high occupancy**, defined as an occupancy rate of at least 80%. We compare two supervised classification approaches: a **baseline logistic regression model**, using a moderate number of structured predictors, and a **high-dimensional logistic LASSO model**, which incorporates a large set of amenity indicators and applies regularization to prevent overfitting.

3.1 Methodology

3.1.1 Baseline Logistic Regression

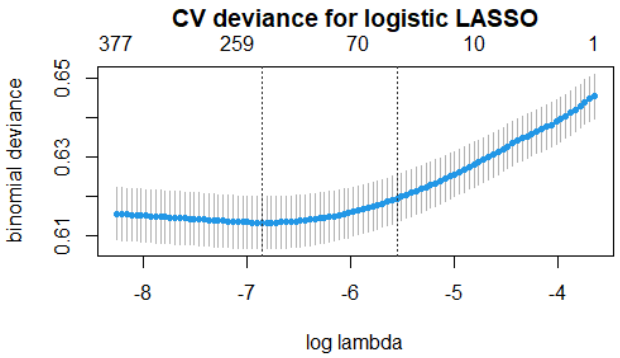
The first model is a standard logistic regression, which estimates the probability that a listing reaches high occupancy as a function of structured predictors. The logistic specification assumes a linear relationship between the predictors and the log-odds of high occupancy. This model is easy to interpret and allows direct economic and managerial insights from the estimated coefficients.

However, logistic regression becomes less effective when the number of predictors grows large, particularly in the presence of correlated or weakly informative variables, such as detailed amenity indicators.

3.1.2 High-Dimensional Logistic LASSO

To address these limitations, we estimate a **logistic LASSO model** that includes all baseline predictors as well as the full set of amenity indicators. LASSO applies an ℓ_1 penalty to the coefficients, shrinking many of them exactly to zero and effectively performing variable selection.

The penalty parameter λ is chosen using cross-validation, balancing predictive accuracy and model complexity. Two common choices are considered: **λ.min**, which minimizes cross-validated deviance, and **λ.1se**, which yields a more parsimonious model whose deviance lies within one standard error of the minimum.



3.2 Experimental Design

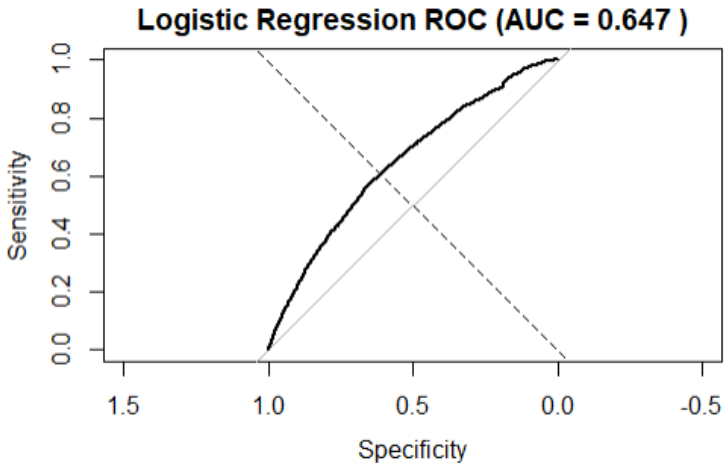
The dataset is randomly split into a **training set (70%)** and a **test set (30%)**, resulting in **50,436 training observations** and **21,615 test observations**. All model estimation is conducted on the training set only. Model performance is evaluated on the test set to ensure an out-of-sample assessment.

Performance is primarily measured using the **Receiver Operating Characteristic (ROC) curve** and the **Area Under the Curve (AUC)**. AUC measures the model's ability to correctly rank listings from low to high booking probability, independently of any specific classification threshold.

3.3 Results

3.3.1 Baseline Logistic Regression Results

The baseline logistic regression achieves an **AUC of 0.647** on the test set, indicating moderate but meaningful predictive power.



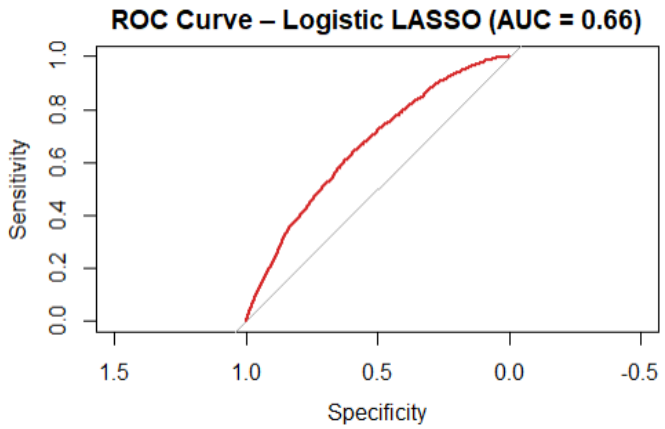
The estimated coefficients provide several important insights. Listings located in **New York and Los Angeles** exhibit significantly higher odds of reaching high occupancy compared to those in Chicago, reflecting stronger demand conditions. **Hotel rooms** perform better than other room types, while **shared rooms** have a strongly negative effect on occupancy probability.

Availability constraints play a critical role: higher values of **minimum nights**, **maximum nights**, and **average available price** significantly reduce the likelihood of high occupancy. This suggests that restrictive booking rules and higher prices discourage sustained demand. Host-related characteristics are also influential. Being a **superhost**, having a **verified identity**, and managing more listings are all associated with higher booking success.

Review-related variables show heterogeneous effects. While overall rating levels are not always statistically significant, **review scores related to value** exhibit a strong positive effect, indicating that perceived value for money is more important than absolute quality ratings.

3.3.2 High-Dimensional Logistic LASSO Results

The logistic LASSO model improves predictive performance, achieving an **AUC of 0.660** on the test set.



Cross-validation selects a penalty parameter that substantially reduces model complexity. Out of **428 potential predictors**, only **42 variables** receive non-zero coefficients. This highlights that most amenities do not independently affect occupancy once location, pricing, and host characteristics are controlled for.

Although λ.min yields the lowest cross-validated deviance, λ.1se produces a more parsimonious model with very similar predictive performance. In applied settings, such a parsimonious specification is often preferred due to greater stability and interpretability.

Using an optimal classification threshold derived from the ROC curve, the LASSO model achieves a balanced tradeoff between sensitivity and specificity.

	Actual	
Predicted	0	1
0	11518	765
1	7964	1368

3.4 Model Comparison

The baseline logistic regression achieves an AUC of **0.6466**, while the high-dimensional LASSO model reaches **0.6603**. Although the improvement is moderate, it is meaningful in large-scale applications where even small gains in ranking accuracy can translate into better pricing or promotion decisions.

Model <chr>	**AUC** <dbl>
Baseline Logistic Regression	0.6465572
High-Dimensional LASSO Logistic	0.6602915

Beyond predictive accuracy, the LASSO results yield substantive insights into the drivers of high occupancy in short-term rental markets. The analysis shows that occupancy is shaped primarily by structural and strategic factors rather than by the sheer number of amenities offered. Room type stands out as the most influential predictor: shared rooms are significantly less likely to achieve high occupancy,

reflecting a strong and persistent preference for privacy that remains robust after controlling for price, location, and host characteristics.

Host-related quality signals also play a central role. Superhost status is consistently associated with higher occupancy, underscoring the importance of trust and platform-generated reputation mechanisms in peer-to-peer markets. Among review-related variables, perceived value emerges as a key predictor, while total ratings are not systematically selected. This suggests that guests prioritize value for money over absolute quality when making booking decisions.

Pricing structure and booking constraints further influence occupancy outcomes. More restrictive rules—such as minimum and maximum night requirements or limitations on long-term stays or pets—are negatively associated with high occupancy, highlighting a clear tradeoff between maximizing occupancy and pursuing alternative revenue or management strategies.

A notable contribution of the LASSO approach lies in its treatment of amenities. Despite starting from a very high-dimensional set, the model retains only a small and highly selective subset of amenities, typically related to convenience or accessibility (e.g., self check-in or child-friendly features). Generic or regulatory amenities do not meaningfully predict occupancy once core listing characteristics are accounted for, indicating that most amenities have limited independent explanatory power.

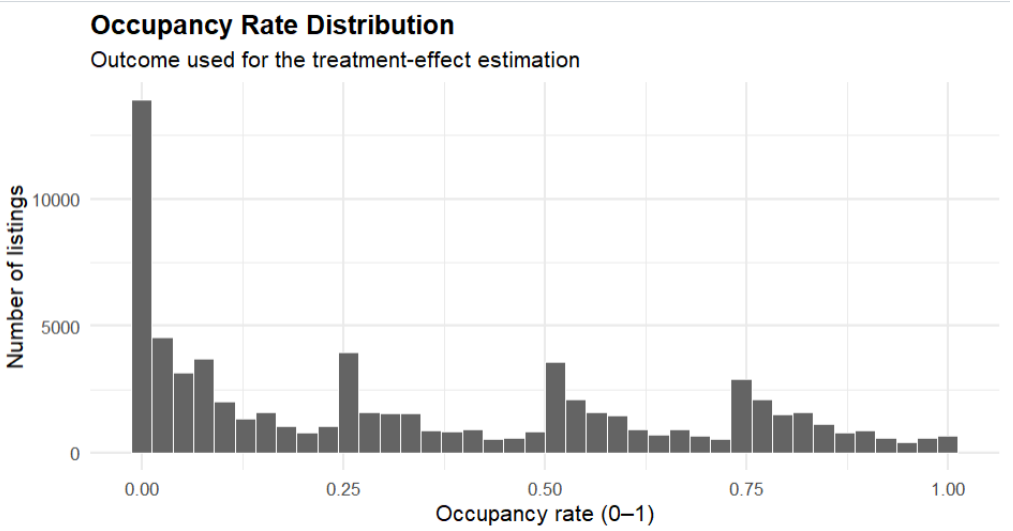
Finally, the moderate AUC values should be interpreted as a feature of the underlying data-generating process rather than a methodological shortcoming. Booking behavior depends on many unobserved and time-varying factors that are not captured in the dataset. Consequently, the model is best viewed as a ranking and decision-support tool rather than a strict classifier.

In conclusion, regularized logistic models provide a scalable and interpretable framework for analyzing occupancy in short-term rental markets. The results highlight the dominant role of room type, location, pricing policies, and host reputation, while showing that only a narrow set of amenities contributes meaningfully to booking success. These findings are economically intuitive and offer actionable insights for both hosts and platform designers.

4. TREATMENT EFFECT - Jaime Martínez

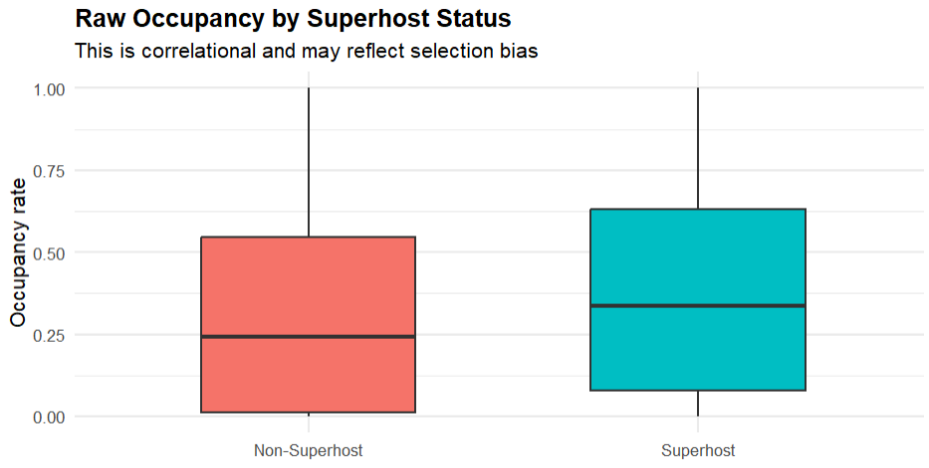
The Superhost badge is one of the most prominent quality signals on Airbnb. Listings managed by Superhosts consistently appear to achieve higher occupancy rates than non-Superhost listings. However, this observed difference may not reflect a causal effect of the badge itself. Superhost status is earned through sustained high performance, including excellent reviews, fast response rates, and low cancellation behavior, rather than being randomly assigned.

As a result, Superhosts are likely to differ systematically from other hosts in ways that directly affect occupancy, such as listing quality, amenities, pricing strategy, experience, and location. The objective of this section is to determine whether Superhost status **causally increases occupancy**, or whether the observed advantage is driven by underlying characteristics of the listings and hosts.



We begin by examining the distribution of the outcome variable used throughout the analysis: the occupancy rate. The histogram shows a highly dispersed and right-skewed distribution, with a large mass of listings exhibiting very low occupancy and a smaller, yet non-negligible, share of listings reaching high occupancy levels close to full booking.

This heterogeneity highlights that occupancy is driven by multiple interacting factors rather than a single dominant characteristic. As a result, any attempt to isolate the effect of a specific feature—such as Superhost status—requires controlling for a broad set of confounding variables that may explain why some listings perform significantly better than others.



A naive comparison of occupancy rates by Superhost status reveals a clear and substantial difference. The boxplot indicates that Superhost listings consistently achieve higher occupancy across the distribution, with both higher medians and higher upper quartiles relative to non-Superhost listings.

This pattern is also reflected in mean values: Superhosts exhibit an average occupancy rate of approximately 38.5%, compared to 30.9% for non-Superhosts, corresponding to a raw difference of 7.62 percentage points. At first glance, this evidence may suggest that the Superhost badge plays an important role in increasing demand.

WHY NAIVE EVIDENCE IS MISLEADING?: SELECTION AND CONFOUNDING

Despite the strong raw association, this evidence cannot be interpreted causally. Superhost status is not randomly assigned; it is earned through sustained performance, high review scores, responsiveness, and low cancellation rates. Consequently, Superhosts are likely to differ from other hosts along many dimensions that directly affect occupancy, such as listing quality, amenities, pricing strategies, experience, and location.

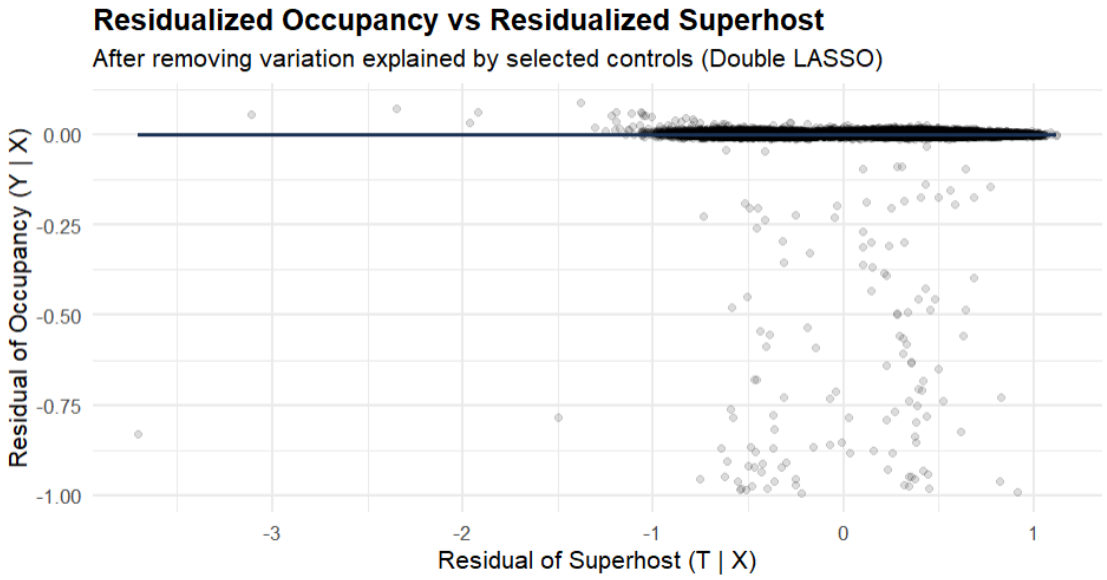
The observed difference in occupancy may therefore reflect selection effects rather than a true causal impact of the badge itself. Ignoring these confounding factors risks attributing to Superhost status what is in fact driven by pre-existing listing and host characteristics.

CAUSAL STRATEGY: INTUITION BEHIND DOUBLE LASSO

To address this challenge, we implement a Double LASSO (partialling-out) approach designed for high-dimensional settings. This method allows us to control for a very large number of potentially correlated covariates—including more than 400 amenity indicators—while avoiding overfitting.

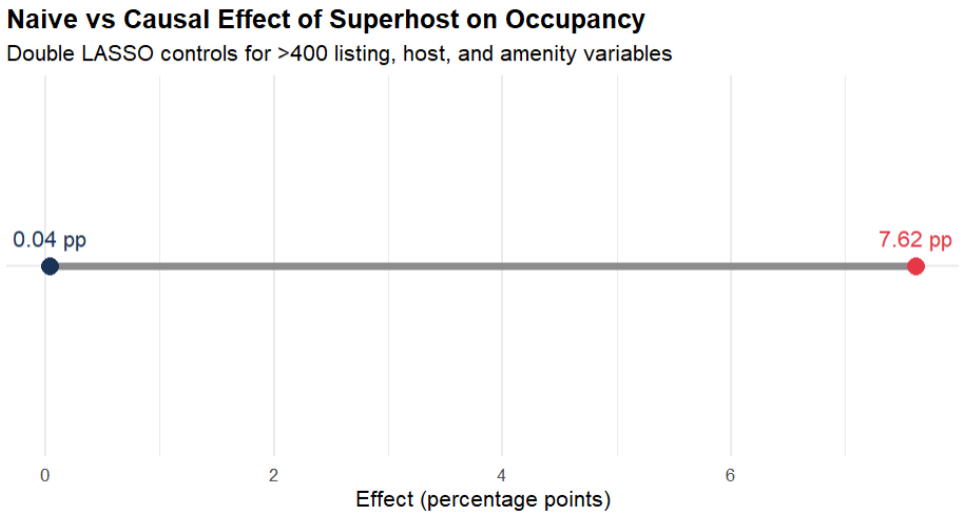
The intuition behind the approach is to isolate the component of Superhost status that is orthogonal to observable listing and host characteristics. By residualizing both occupancy and Superhost status with respect to the union of variables selected in two separate LASSO models, we obtain variation in Superhost status that is plausibly exogenous, conditional on observables.

INTERPRETATION OF THE RESIDUALIZED REGRESSION



The causal estimates obtained from the residualized regression indicate that Superhost status has an extremely small and statistically insignificant effect on occupancy. The estimated effect is approximately 0.04 percentage points, with a p-value of 0.15 and a robust confidence interval that includes zero.

The residualized scatter plot further reinforces this conclusion. Once the variation explained by selected controls is removed, the relationship between Superhost status and occupancy becomes nearly flat, suggesting that Superhost status does not independently drive booking intensity.



The contrast between naive and causal estimates is particularly striking when visualized directly. While the raw comparison suggests a sizable positive effect of more than seven percentage points, the Double LASSO estimate collapses this effect to nearly zero once controls are introduced.

This figure succinctly summarizes the core finding of the analysis: the apparent advantage of Superhosts in the raw data is almost entirely explained by observable characteristics rather than by the badge itself. The Superhost designation does not generate additional demand once listing quality and host behavior are taken into account.

Taken together, the evidence strongly suggests that Superhost status does not causally increase occupancy. Although Superhosts appear to perform better in raw data, this advantage disappears once we control for listing characteristics, amenities, pricing, and host professionalism using a high-dimensional causal framework.

The Superhost badge should therefore be interpreted primarily as a signal of existing quality rather than as a mechanism that actively boosts bookings. Hosts do not become more successful because they obtain the badge; instead, they obtain the badge because they are already operating high-quality listings.

More broadly, these results highlight the importance of distinguishing correlation from causation in platform-based markets. Features that appear powerful in descriptive analyses may lose their apparent effectiveness once selection and confounding are properly addressed. From a policy perspective, improving listing fundamentals may be far more impactful than badge-based incentives alone.

CONCLUSION

This project uses large-scale Airbnb data from various U.S. cities to study how listing features relate to pricing, demand, and market structure. A consistent theme emerges: Airbnb listings exist in a highly diverse market. Outcomes are shaped by a mix of structural features, host behavior, and demand dynamics rather than a single major factor.

The segmentation analysis shows that listings naturally group into understandable market segments. We use both amenity-based and performance-based approaches to identify systematic differences in what listings provide and how they perform regarding price, quality signals, and occupancy. This highlights that Airbnb is not a uniform marketplace but a collection of unique sub-markets with varying performance profiles.

Our pricing analysis shows that nightly prices mainly depend on structural and location-related characteristics. These include property size, room type, and neighborhood effects. Regularization methods reveal that while many variables exist, only a small subset consistently explains price variation. Softer signals like reviews and host traits play a more limited role. This states that pricing is more about listing fundamentals than short-term demand signals.

The occupancy modeling adds to this understanding by focusing on booking behavior instead of price levels. By distinguishing highly booked listings from others, the analysis shows that demand is linked to a different set of features compared to price alone. Availability patterns, review activity, and host engagement are key. This reinforces the idea that pricing power and booking intensity, while related, are distinct parts of listing performance.

Finally, the treatment-effect analysis on Superhost status examines the causal aspect of host quality. After accounting for observable listing and host traits, the apparent association between Superhost designation and occupancy largely vanishes, indicating that much of the raw performance difference is driven by underlying host and listing attributes rather than the badge itself. This result highlights the importance of distinguishing correlation from causation when interpreting platform-generated quality signals.

Overall, the project offers a unified view of Airbnb performance across segmentation, pricing, demand, and host status. The findings show how we can use efficient computing techniques along with modern statistical and machine learning tools to extract clear, useful insights from large and complex datasets. Practically, the results guide hosts in benchmarking their listings and prioritizing improvements. They also illustrate broader market patterns relevant to platform design and policy analysis.

FUTURE WORK & IMPROVEMENTS

While the analysis provides clear insights into pricing, demand, segmentation, and host quality, several limitations suggest natural directions for future work. Many of the extensions discussed below were not pursued in the current project due to space and scope constraints, rather than conceptual or methodological limitations.

First, the segmentation analysis is performed on pooled data across cities, which implicitly assumes that market structure is comparable across locations. A useful extension would be to examine segmentation stability at the city level or allow cluster structures to vary by market, capturing differences in local demand, housing stock, and competition intensity.

Second, segmentation could be more tightly integrated with the pricing and occupancy models. Rather than treating clusters as descriptive outcomes, future work could incorporate segment membership directly as explanatory variables in the price and occupancy regressions, allowing for segment-specific pricing elasticities or demand responses. This would strengthen the link between market structure and performance outcomes.

For the Superhost treatment-effect analysis, the current approach relies on observed covariates and cross-sectional variation. Future work could exploit panel structure more fully, tracking listings over time as hosts gain or lose Superhost status, or incorporating additional controls for time-varying demand shocks. This would further strengthen causal interpretation and reduce reliance on selection-on-observables assumptions.